

Different Cluster-Loading Protocols Have No Effect on Intraset and Interset Power Expression

Charles E.P. Dudley, Eric J. Drinkwater, and Simon A. Feros

Center for Sport Research (CSR), School of Exercise and Nutrition Sciences, Deakin University, Geelong, Australia

Abstract

Dudley, CE, Drinkwater, EJ, and Feros, SA. Different cluster-loading protocols have no effect on intraset and interset power expression. *J Strength Cond Res* XX(X): 000–000, 2020—This study investigated the effect of cluster-loading protocols on changes in hang power clean (HPC) power characteristics. Ten trained weightlifters (7 male, 3 female, HPC 1.28 [M]; 1.03 [F] × body mass) completed 4 experimental protocols in a randomized repeated-measures design. Protocols were traditional, a basic cluster, and 2 undulating clusters. Each protocol comprised 3 sets of 5 HPC repetitions at an average of 70% 1 repetition maximum (RM). The load for undulating cluster protocols ranged between 60 and 80% 1RM. Countermovement jumps (CMJs) were performed before and after each protocol to gauge effect on neuromuscular status. Statistical analysis comprised a one-way repeated measures analysis of variance. Peak power was significantly greater in the fifth repetition of the basic cluster set protocol compared with the traditional set protocol (mean ± SD: $6.6 \pm 2.8\%$, $p = 0.043$, $g = 0.25$). Neither undulating load protocol significantly altered set power characteristics compared with basic cluster sets; however, when analyzing repetition data, peak power increased as load increased (and vice versa), indicating the prescribed loads were suboptimal for power expression in this sample. There was no change in CMJ height between protocols. Coaches should prescribe inter-repetition rest to maintain peak power across multiple repetitions and sets. Basic cluster protocols seem to be the most practical choice over undulating cluster protocols because there is no need to adjust load between repetitions.

Key Words: inter-repetition rest, resistance training, weightlifting, hang power clean, undulation

Introduction

Muscular power is a key physical capacity that relates to sports performance in a number of sports, particularly those where force production in a short period of time is required, such as weightlifting (21). When following a well-structured resistance training program, muscular power can be enhanced through both neural and muscular adaptations (27). Cluster set training is an advanced training method that can be used to maintain acute muscular power throughout a set, facilitated through the maintenance of velocity (19,29). Cluster set training is a resistance training method whereby set structure is altered so rest periods are taken between a group of repetitions within a set (a cluster), contrasting traditional sets where repetitions are performed in a continuous manner (29). By interspersing rest within a set, cluster sets have previously been shown to maintain peak power during a set, whereas power may decrease in traditional training (19). Training with high power outputs may have positive implications for long-term adaptation in power-based training programs (9).

When programming a cluster set protocol, there are 2 common ways of distributing rest. These 2 methods either add inter-repetition rest additional to interset rest or redistribute interset rest between repetitions (29). The “additional rest cluster set” protocol has limited use in research as any change in biomechanical variables observed may be because of one of 2 altered independent variables: increased total rest and/or the redistribution of rest (29). By extending the total time taken to

complete a single exercise, the additional rest cluster set protocol also increases the opportunity cost of programming that exercise, with some cluster set protocols taking up to 10 minutes longer per exercise than traditional set training (13). Conversely, the rest redistribution protocol, which maintains total rest by decreasing interset rest and increasing inter-repetition rest, ensures that any changes to dependent variables are because of rest redistribution as opposed to additional rest.

Undulating load is another cluster set method that has been previously believed to alter power output; however, the effect of undulating load is largely unclear (12). Load would normally be undulated across sets; however, because of the nature of a cluster set, load can be manipulated within a set. Haff et al. (12), explored the effect of undulating-load cluster sets on the clean pull exercise performed at an average of 90 and 120% of 1 repetition maximum (RM) power clean in one set of 5 repetitions. Cluster sets resulted in an increase in bar velocity in both the 90 and 120% protocols compared with traditional sets, with no difference for undulating load (12). Despite the lack of significant findings pertaining to the undulation of load by Haff et al. (12), the efficacy of alternate undulating load protocols using different exercises has not been explored. Exploration of these protocols within cluster set training is warranted to further inform coaches about best-practice for maintaining power output across multiple sets and repetitions.

The purpose of this study was to assess the effects of manipulating cluster set loading strategies on acute biomechanical variables in the hang power clean (HPC) in trained weightlifters. Specifically, this study will explore whether undulating load within a set will affect peak power, peak force, and peak velocity. It is our hypothesis that basic cluster sets will maintain peak power, and

Address correspondence to Charles E.P. Dudley, Charles.dudley@outlook.com.au.

Journal of Strength and Conditioning Research 00(00)/1–7

© 2020 National Strength and Conditioning Association

peak velocity to a greater extent than traditional sets; however, there will be no difference in peak force. This is because of changes in peak power being driven by changes in velocity (23); however, peak power, peak velocity, and peak force will be analyzed to fully understand the kinetic and kinematic differences between protocols. Furthermore, it is hypothesized that the undulating load protocols will result in altered peak power, peak force, and peak velocity when compared with the basic cluster sets.

Methods

Experimental Approach to the Problem

This research implemented a cross-sectional, repeated measures study design. Subjects completed 6 total training sessions; one session per week for 6 weeks. Sessions were conducted at the same time of the day to control for diurnal variation. The first 2 sessions involved determination of 1RM HPC, and a traditional set protocol to establish test-retest, inter-rep, and inter-set reliability. The remaining 4 sessions, conducted in randomized order, were: (a) traditional sets, (b) basic cluster sets, (c) high-low-high (HLH) undulating cluster sets, and (d) low-high-low (LHL) undulating cluster sets. Three counter-movement jumps (CMJs) were performed after the specific warm-up and at the end of each protocol to ascertain the effects of each protocol on subsequent neuromuscular status (5).

Dependent variables were collected using a linear position transducer (GymAware Powertool; Kinetic Performance Technology, Canberra, Australia) and included: (a) absolute peak power, (b) absolute peak force, and (c) absolute peak velocity. Peak force and peak velocity have been included to provide a holistic description of the movement kinetics and kinematics. The dependent variable for the CMJ was jump height. Independent variables in this study included: (a) rest distribution, and (b) load distribution. For the purpose of this investigation, the traditional set and basic cluster set protocols were compared to test the effects of rest distribution on movement kinetics and kinematics. The basic cluster set was compared with the 2 undulating cluster set protocols to examine the effects of load distribution on movement kinetics and kinematics.

Subjects

Ten weightlifters (male = 7; female = 3) experienced in performing the HPC participated in this study (Table 1). Subjects were all in a noncompetition phase of training and were asked to refrain from strenuous physical activity for 48 hours before testing, and to maintain a similar training volume throughout the study. However, no information was gathered regarding the subject's specific phase of training or current training loads. All subjects met the following inclusion criteria: between 18 and 35 years of age; >2 years' experience in weightlifting; self-reported Sinclair coefficient >160 for women and >230 for men (1); no recent (6 months) musculoskeletal injury that may affect performance of the HPC, as indicated by the subject. Self-reported Sinclair was used because of reasons such as excessive time between competitions. A priori statistical power analysis indicated 10 subjects were required to yield a statistical power of 80% ($\alpha = 0.05$; effect size = 0.1); (G*Power version 3.1.7, Institute of Experimental Psychology, Heinrich Heine University, Dusseldorf, Germany (7)). All procedures were approved by the university's Human Research Ethics Committee (HEAG 27-2018). Written informed consent was obtained for each subject before the commencement of the study.

Table 1
Subjects characteristics.*†

Variables	Male	Female
No. of subjects	7	3
Age (yrs)	25.8 ± 3.1	26.3 ± 1.1
Height (cm)	178.7 ± 7.6	167.0 ± 6.9
Weight (kg)	88.4 ± 16.5	68.2 ± 8.0
Sinclair coefficient	285.4 ± 40.3	190.3 ± 15.0
Absolute hang power clean 1RM (kg)	112.1 ± 15.2	70.8 ± 10.1
Relative hang power clean 1RM (kg·BW ⁻¹)	1.28 ± 0.18	1.04 ± 0.29

*1RM = 1 repetition maximum.

†Data are mean ± SD.

Procedures

In the first session, subjects completed 1RM testing and were measured for standing height and weight collected using a SECA 213 portable Stadiometer (SECA, Hamburg, Germany) and UC-321 Series Precision Scales (A&D Weighing, Melbourne, Australia), respectively. Men and women completed all protocols using 20- and 15-kg calibrated barbells respectively (Eleiko, Halmstad, Sweden; Uesaka, Tokyo, Japan). All conditions were preceded by a warm-up that included dynamic stretching exercises of the quadriceps, hamstrings, adductors, quadriceps, and gastrocnemius (31). To establish a 1RM, subjects estimated their HPC 1RM based off previous performance and then performed a series of 5 sub-maximal sets of one to 2 repetitions between 50 and 90% 1RM (10). After this, the subjects made 2.5–5 kg increases in weight to establish a point of failure, dependent on the perceived difficulty of the completed repetition, resting approximately 3 minutes between each set, as per previously published 1RM testing protocol (10). No subjects completed more than 3 maximal attempts, and once the subject reached the point of failure, the most recently completed attempt was recorded as the 1RM. Failure was classified as an inability to execute the “catch” phase of the lift or going below parallel in the recovery phase of the movement, judged by the lead investigator (C.D.), an accredited coach with the Australian Weightlifting Federation (18).

During the experimental conditions, subjects then completed a specific warm-up consisting of 35% 1RM for 5 repetitions, 50% 1RM for 3 repetitions, and one repetition at the load equal to that of the first working set. Three minutes after the specific warm-up, subjects completed CMJ testing, which comprised 3 maximal-effort jumps with a 1-minute recovery between trials (5). Subjects were instructed to stand with a wooden dowel placed on their upper back in a position similar to the back squat. Subjects were instructed to limit dowel movement, self-select countermovement depth, jump as high as possible, and to land in the same spot, following prior published protocols and manufacturer recommendations (23). The linear position transducer was attached to the left-hand side of the dowel for all trials. Jump height was averaged across 3 trials to improve test-retest reliability (5). Jump height was recorded by the manufacturer's software as the change in vertical displacement from the initiation of the jump to the point of peak positive displacement (32). The linear positions transducer has previously been shown to overestimate jump height, however has displayed good reliability, therefore is suitable for assessing change in jump height (32). A further 3 CMJs were completed 3 minutes after the experimental protocol had finished. The length of the rest between the conditioning activity and the CMJ was selected as cluster sets have been shown to decrease the time required to exhibit potentiation after a conditioning activity (4).

All working sets were completed at an average load of 70% 1RM, as previous research has indicated that this is within the optimal range for peak power output in the HPC (15). Undulating cluster sets were conducted in the range of 60–80% 1RM. The LHL cluster set comprised an undulation of load throughout the set of 65% 1RM, 70% 1RM, 80% 1RM, 70% 1RM, and 65% 1RM. The HLH cluster set involved an undulation of load throughout the set of 75% 1RM, 70% 1RM, 60% 1RM, 70% 1RM, and 75% 1RM. Thirty seconds of inter-repetition rest was prescribed as previous research into cluster training has generally been conducted using 20–40-second inter-repetition rest (19); however, no defined protocols have been developed. Equalized total rest time was used between the conditions (totalling 720 seconds) to ensure the change in dependent variables in the traditional and basic cluster set protocols was only because of the redistribution of rest. The distribution of load and rest is illustrated in Figure 1.

No technical cueing or encouragement was given throughout the experimental protocols; however, encouragement and basic technical cueing was consistently provided during 1RM testing. A standardized protocol regarding athlete instructions was used throughout the testing procedure, as athlete instruction has the potential to influence the performance of the jump (33). Before the commencement of a lift, the athlete was given the instruction “Please bring the bar to your hip; 3-2-1 GO!”

Statistical Analyses

Raw data were extracted from the GymAware Pro online portal to Microsoft Excel 2016 (Microsoft Corporation, Redmond, WA). Data cleaning was conducted before statistical analysis to ensure that all values were plausible and there were no missing results. Outliers in the data were assessed by visual inspection of the box and whisker plot. If an outlier was identified, the datum was then visually inspected to determine the reason for the outlier. No outliers were removed because they were deemed to be plausible values. All data were normally distributed as assessed by the Shapiro-Wilk test ($p \geq 0.05$) and box and whisker plots. Mauchly’s test of sphericity indicated that the assumption of sphericity had not been violated for most variables ($p \geq 0.05$). In the event that the assumption of sphericity was violated ($p \leq 0.05$), the Greenhouse-Geisser correction was applied.

The analyses involved investigating the inter-repetition data and inter-set data. Inter-repetition analysis involved calculation of the percentage change from the first repetition (sets pooled) for each set

protocol, whereas inter-set analysis comprised the absolute data to calculate the average across each set (repetitions pooled) for each set protocol. This analytical approach has been used previously in cluster set research, and was conducted to best illustrate the effect of cluster sets on the change in biomechanical variables within a set, and the overall change across multiple sets (13,23).

To analyze the effects of rest distribution within and across cluster sets, the traditional and basic cluster protocols were compared with a three-way repeated measures analysis of variance (ANOVA). Three factors were entered for this analysis: PROTOCOL (2 levels: traditional, basic cluster), SET (3 levels: set 1, set 2, set 3), and REPETITION (5 levels: repetition 1, repetition 2, repetition 3, repetition 4, repetition 5). The basic cluster protocols were compared with the 2 undulating cluster protocols with a three-way repeated measures ANOVA to determine the effects of load distribution within and across cluster sets. Three factors were entered for this analysis: PROTOCOL (3 levels: basic cluster, HLH undulating cluster, LHL undulating cluster), SET (3 levels: set 1, set 2, set 3), and REPETITION (5 levels: repetition 1, repetition 2, repetition 3, repetition 4, repetition 5). For the inter-repetition analysis in this component, the repetition data from the undulating protocols was calculated as the percentage change in relation to each repetition of the basic cluster protocol, and the data for the basic cluster was therefore set to 0 for each repetition. This was conducted to facilitate easier comparison of the 2 undulating protocols to the basic cluster protocol, which served as the “control” protocol in this analysis. The 4 protocols were compared with a two-way repeated measures ANOVA to ascertain differences between protocols with respect to neuromuscular status. Two factors were entered for this analysis: PROTOCOL (4 levels: traditional, basic cluster, HLH undulating cluster, LHL undulating cluster), and TIME (2 levels: pre, post). Where a statistically significant interaction existed for the repeated measures ANOVA, Fisher’s Least Significant Difference post-hoc analyses were conducted to explore relevant and planned pairwise differences within and between the protocols (16). Effect sizes were calculated for each significant pairwise difference located within and between protocols. To avoid the positive bias associated with Cohen’s d in small sample sizes, Hedges’ g statistic was computed in a purpose made excel spreadsheet (6).

Absolute reliability of all variables was assessed by the standard error of measurement (SEM), whereas relative reliability was determined via log-transformed coefficient of variation (expressed as a percentage) and the intraclass correlation

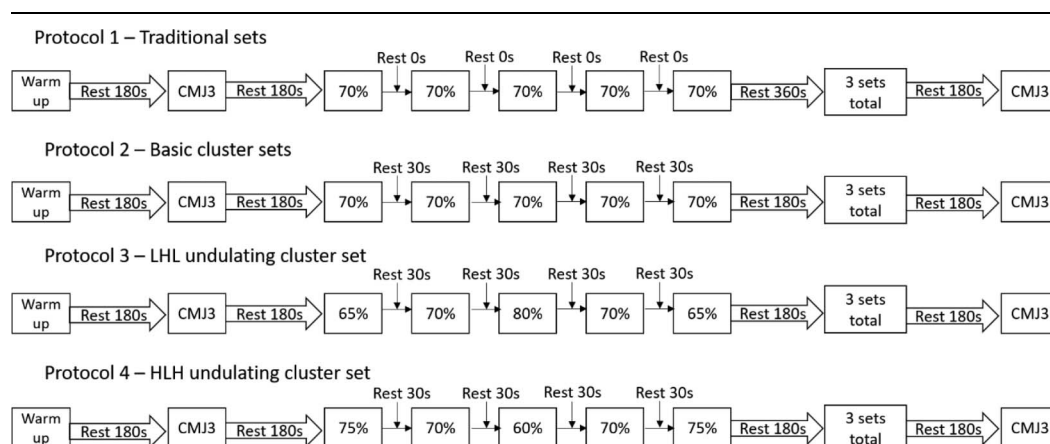


Figure 1. Traditional and cluster loading protocols. LHL = low-high-low; HLH = high-low-high; CMJ3 = countermovement jump protocol comprising 3 jumps.

coefficient (model_{2,k}). Reliability was calculated as: (a) test-retest (i.e., traditional protocol 1 vs. traditional protocol 2), (2) inter-rep (i.e., comparison of repetitions 1–5 within traditional protocol 1), and (3) inter-set (i.e., comparison of sets 1–3 within traditional protocol 1) (Table 2). All reliability data were calculated in a purpose-made excel spreadsheet (14).

All statistical analyses were performed in IBM SPSS Statistics (Version 25, IBM Corp., Armonk, NY), with all data preparation performed in Microsoft Excel 2016 (Microsoft Corporation). Statistical significance was set at $p \leq 0.05$ for all analyses.

Results

Traditional vs. Basic Cluster Protocols

There was a significant $\text{PROTOCOL} \times \text{REPETITION}$ interaction for peak power, $F_{(4, 36)} = 3.002$, $p = 0.031$. Peak power was significantly lower in the fifth repetition in the traditional protocol compared with the basic cluster protocol (mean $\pm$ SD: $6.6 \pm 2.8\%$, $p = 0.043$, $g = 0.25$; Figure 2).

There was a significant $\text{PROTOCOL} \times \text{SET}$ interaction for peak power, $F_{(2, 18)} = 5.374$, $p = 0.015$; however, post-hoc analyses did not detect any statistically significant between-protocol differences.

There was a significant $\text{PROTOCOL} \times \text{REPETITION}$ interaction for peak velocity, $F_{(4, 36)} = 3.072$, $p = 0.028$ and peak force, $F_{(4, 36)} = 3.246$, $p = 0.023$; however, post-hoc analyses did not detect any statistically significant between-protocol differences.

Basic Cluster vs. Undulating Cluster Protocols

There was a significant $\text{PROTOCOL} \times \text{REPETITION}$ interaction for peak power, $F_{(8, 72)} = 3.458$, $p = 0.002$. Peak power was significantly higher in the third repetition of the LHL cluster protocol compared with the basic cluster protocol (mean $\pm$ SD: $5.4 \pm 1.1\%$, $p = 0.001$, $g = 0.24$, Figure 3). Peak power was significantly higher in the fifth repetition of the HLH cluster protocol compared to the basic cluster protocol (mean $\pm$ SD: $4.2 \pm 1.8\%$, $p = 0.046$, $g = 0.19$, Figure 3).

There was a significant $\text{PROTOCOL} \times \text{REPETITION}$ interaction for peak velocity, $F_{(8, 72)} = 27.418$, $p < 0.001$. Significant differences in peak velocity were detected between the basic cluster and LHL cluster protocols in the first repetition (mean $\pm$ SD: $3.5 \pm 1.4\%$, $p = 0.032$, $g = -0.47$), third repetition (mean $\pm$ SD: $-3.8 \pm 1.2\%$, $p = 0.014$, $g = 0.67$), and fifth repetition (mean $\pm$ SD: $4.9 \pm 1.3\%$, $p = 0.004$, $g = -0.66$) (Figure 4). Significant differences in peak velocity were observed between the basic cluster and HLH cluster protocols in the first repetition (mean $\pm$ SD: $-4.4 \pm 1.3\%$, $p = 0.008$, $g = 0.7$) and third repetition (mean $\pm$ SD: $8.2 \pm 1.0\%$, $p < 0.001$, $g = -1.36$) (Figure 4).

There was a significant $\text{PROTOCOL} \times \text{REPETITION}$ interaction for peak force, $F_{(2, 72)} = 22.036$, $p < 0.001$. Significant differences in peak force were detected between the LHL cluster protocols in the first repetition (mean $\pm$ SD: $-6.3 \pm 1.4\%$, $p = 0.002$, $g = 0.14$), third repetition (mean $\pm$ SD: $6.8 \pm 1.6\%$, $p = 0.002$, $g = -0.53$), and fifth repetition (mean $\pm$ SD: $-7.23 \pm 1.4\%$, $p < 0.001$, $g = 0.07$) (Figure 5). Significant differences in peak force were detected between the basic cluster and HLH cluster protocols in the first repetition (mean $\pm$ SD: $7.1 \pm 1.6\%$, $p = 0.002$, $g = -0.33$), third repetition (mean $\pm$ SD: $-5.9 \pm 1.4\%$, $p = 0.002$, $g = 0.28$), and fifth repetition (mean $\pm$ SD: $-8.4 \pm 1.6\%$, $p = 0.001$, $g = -0.40$) (Figure 5).

Table 2
Reliability data.*†

	SEM	CV	ICC
Test-retest			
Peak power	86.56 (59.54–158.03)	3.5 (2.4–6.4)	0.98 (0.93–1.0)
Peak velocity	0.04 (0.02–0.06)	1.8 (1.2–3.3)	0.94 (0.77–0.98)
Peak force	51.08 (35.14–93.26)	3.5 (2.4–6.4)	0.98 (0.92–1.0)
Inter-rep			
Peak power	62.77 (49.88–85.01)	2.9 (2.3–3.9)	0.99 (0.97–1.0)
Peak velocity	0.04 (0.03–0.06)	2.1 (1.7–2.9)	0.92 (0.81–0.98)
Peak force	35.79 (28.45–48.48)	2.8 (2.2–3.8)	0.99 (0.97–1.0)
Inter-set			
Peak power	41.53 (31.97–60.71)	2.0 (1.5–2.9)	0.99 (0.98–1.0)
Peak velocity	0.02 (0.02–0.03)	1.1 (0.8–1.8)	0.98 (0.93–0.99)
Peak force	32.86 (24.09–51.89)	2.3 (1.7–3.7)	0.99 (0.97–1.0)

*ICC = intraclass correlation; CV = coefficient of variation.

†Data are mean ($\pm$ 95% confidence interval).

Countermovement jump Performance

There was no statistically significant $\text{PROTOCOL} \times \text{TIME}$ interaction for jump height in the CMJ.

Discussion

The purpose of this study was to assess the effects of manipulating cluster set loading strategies on acute biomechanical variables in the HPC in trained weightlifters. It was hypothesized that basic cluster sets will maintain peak power, and peak velocity to a greater extent than traditional sets and that the undulating load protocols will result in altered peak power, peak force, and peak velocity compared with the basic cluster sets. This study found that basic cluster sets are effective in maintaining peak power within a set when compared with a traditional set. Further analysis found that the manipulation of load in undulating protocols did not significantly change absolute peak power over the entire set when compared with basic cluster sets, despite increases in velocity when bar load was reduced and increases in force when

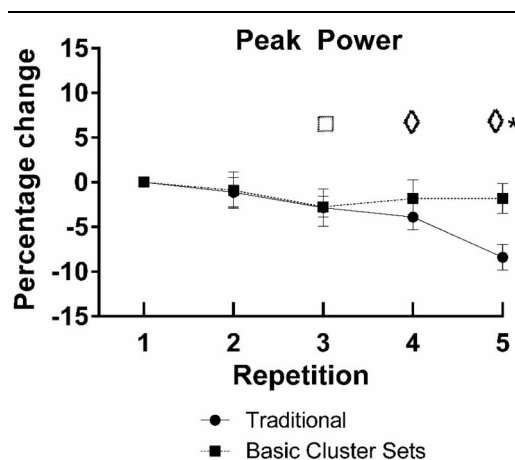


Figure 2. Mean $\pm$ 95% CI peak power (W) percentage change from repetition one across repetitions (sets pooled). *Significant difference between traditional and basic cluster sets ($p < 0.05$). ♦Significant difference from repetition 1 traditional sets ($p < 0.05$). □Significant difference from repetition 1 in basic cluster sets ($p < 0.05$).

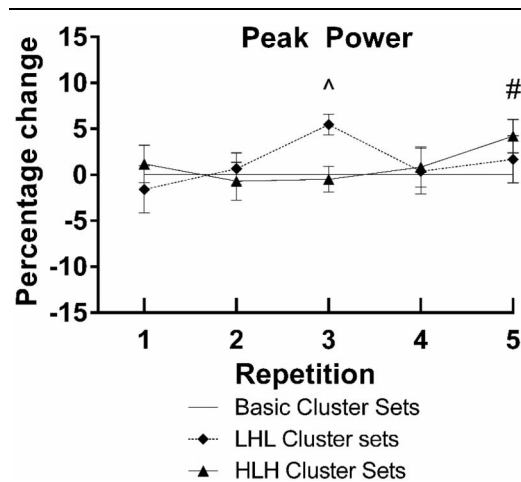


Figure 3. Mean $\pm$ 95% CI peak power (W) percentage change from repetition one across repetitions (sets pooled). ^Significant difference between basic and LHL cluster sets ($p < 0.05$). #Significant difference between basic and HLH cluster sets ($p < 0.05$). LHL = low-high-low; HLH = high-low-high.

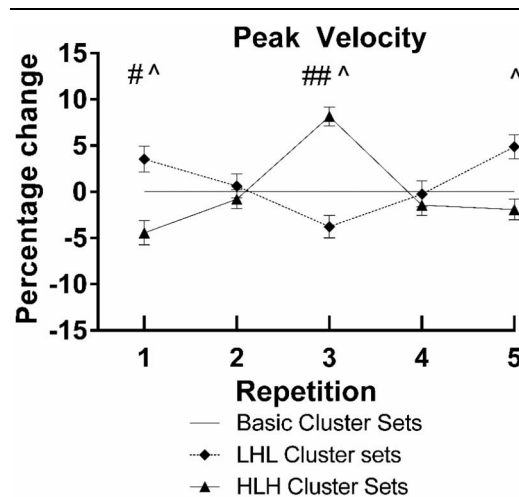


Figure 4. Mean $\pm$ 95% CI peak velocity ($\text{m}\cdot\text{s}^{-1}$) percentage change from repetition one across repetitions (sets pooled). ^Significant difference between basic and LHL cluster sets ($p < 0.05$). #Significant difference between basic and HLH cluster sets ($p < 0.05$; ## = $p < 0.001$). LHL = low-high-low; HLH = high-low-high.

bar load was increased. This demonstrates that differing load configurations during cluster sets do not meaningfully alter power across multiple sets.

To the authors knowledge, this study is the only cluster set study using a weightlifting derivative with equalized total rest time (29). This study provides additional knowledge in cluster set literature; that the change in peak power outputs observed in cluster sets are because the redistribution of rest, and not necessarily the addition of rest. This study represents the only current cluster set investigation using trained weightlifters and equalized total rest time (29). Furthermore, the sample was well trained in comparison to other studies, although the use of different exercises across studies makes direct comparison difficult (29). The relative strength for men within the group was $\sim 30\%$ greater than the relative strength in the HPC reported in Australian Rules Football and Rugby Union players (11,15). A strong rationale for the use of a well-trained population is that power training is often only appropriate once an adequate base level of strength has been built; therefore recruiting trained subjects increases the reliability of the research to the athletes whom the findings may be applicable (27). No cluster set studies have directly examined the HPC, nor have any studies reported the Sinclair coefficient scores of their subjects, making comparison of the relative levels of strength difficult (29).

It was found that the traditional set protocol resulted in a loss of both peak power and peak velocity across repetitions, whereas these variables were maintained within the cluster sets. These findings support the large body of evidence that indicates that cluster sets maintain peak power output (29). The findings also indicate that the maintenance of peak power is due to the maintenance of peak velocity, as opposed to changes in force. This study confirms the findings of Haff et al. (12), that there was no effect of undulating load within cluster sets for absolute peak power.

No effect was found for jump height between protocols. In addition, no change in jump height was found between pre and post conditions. This may be because of a number of factors such as the time between the conditioning activity and the

CMJ, and the intensity of the conditioning activity (26). It has previously been shown that stronger athletes will express greater potentiation when the conditioning activity is closer to an RM, as such the submaximal load used may have effected the magnitude of potentiation (26). Furthermore, the time between the conditioning activity and the CMJ was 3 minutes, whereas previous reviews into the optimal time frame for potentiation indicates that a 7–10 minutes rest should be used (28). The timeframe selected was used because of previous research indicating that the temporal profile for potentiation may be altered by factors such as set structure and strength of the athlete (4,25). Cluster sets are believed to reduce the timeframe for potentiation to be observed because of the decrease in fatigue accumulated compared with traditional sets

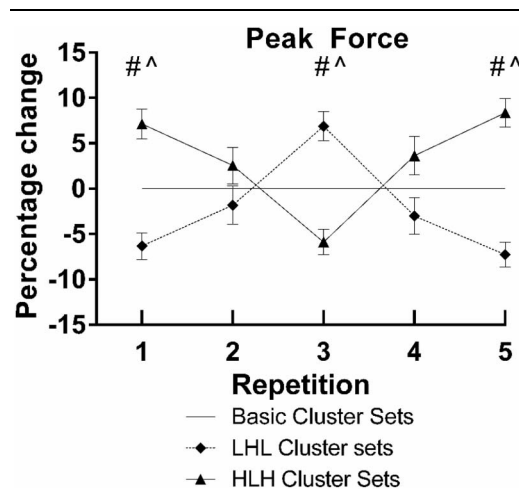


Figure 5. Mean $\pm$ 95% CI peak force (N) percentage change from repetition one across repetitions (sets pooled). ^Significant difference between basic and LHL cluster sets ($p < 0.05$). #Significant difference between basic and HLH cluster sets ($p < 0.05$). LHL = low-high-low; HLH = high-low-high.

(4). Furthermore, the decreased timeline to express potentiation in stronger athletes is believed to be because of the increased fatigue resistance to heavy loads as compared with weaker athletes (25).

This study found that there was no significant effect for peak power or peak velocity when comparing the results between protocols by set. These findings are in contrast to some (30), but not other previous findings (22). The discrepancy in the results between these 3 studies may be explained by the structure of the cluster set. This study, and the study by Merrigan et al. (22), redistributed rest between repetitions, which made total rest time between protocols identical. However, in the study by Tufano et al. (30), inter-repetition rest was added to the inter-set rest; meaning the cluster set protocols had additional total rest compared with the traditional set. No study has directly compared these 2 methods of adding inter-repetition rest.

The results of the undulating cluster sets indicate the optimal load for producing peak power output in this sample was $\geq 80\%$ 1RM. This can be identified by the trend shown in Figure 3 that any adjustment of load away from 80% 1RM resulted in a decrease in peak power output. The average intensity of all sets was set at 70% , based on the findings of previous research which indicated optimal load was likely to be between 70 and 80% 1RM (15,17). Average load was set at 70% because of further analysis by Kawamori et al. (15), who proposed optimal load in stronger athletes to be 70% . A possible reason for the discrepancy in optimal load was that the current study used a linear position transducer to calculate velocity of the barbell. The power and force outputs were then calculated based on the barbell's velocity, using an inverse dynamics approach (8). Conversely, the studies by Kawamori et al. (15), and Kilduff et al. (17), used force platforms and calculated "system" power using the forward dynamics approach (20). Previously, McBride et al. (20), has shown that using the "system" approach can under-estimate optimal load by 10% when compared with optimal load measured by the barbell. Given that the current study measured bar peak power, the elevated proposed optimal load may be explained by the oversight in the initial study design of the different methodology. The use of a suboptimal load is a primary limitation of this study.

Although the calculation of power from barbell velocity may be observed as an inherent weakness of the current study, previous literature indicates that calculating barbell velocity may be more appropriate for weightlifters (20). Furthermore, measuring the velocity of the barbell alone is often more practical, with a number of portable, affordable, valid, and reliable velocity devices available, whereas even portable force platforms are expensive and cumbersome (3,24). However, a limitation of this study is that barbell velocity was calculated with a single linear position transducer, as opposed to a gold standard system such as 3D motion capture (2). Given the availability of portable velocity devices, literature may be best served to focus on reporting optimal loads as a function of barbell velocity to allow the data to be relatable to coaches. This research further highlights the importance of using optimal loads that are specific to the athlete and ensuring the consistency of method for measuring optimal load.

This study shows that there is no effect of manipulating load on peak power in the cluster set, and no effect of loading protocol on acute neuromuscular performance. However, the results on undulating load were compromised by a less than optimal prescribed load. This study supported that cluster sets maintain peak power and peak velocity across sets and repetitions when

compared with traditional set, even in the absence of additional rest. Future research should be conducted whereby optimal load for subjects was individualized.

Practical Applications

The ability of cluster sets to maintain peak power and peak velocity will have practical implications for coaches who are trying to maintain power outputs, while also maintaining training volume. A coach who is prescribing multiple sets of at least 3 repetitions may be best served to also prescribe decreased inter-set rest and increased inter-repetition rest to maintain peak power. We recommend that basic cluster sets over traditional sets be used because of the maintenance of peak power. Furthermore, traditional cluster sets are recommended over undulating cluster sets because of their equivalence in power, but increased simplicity.

Acknowledgments

The authors would like to thank all subjects for volunteering their time for this study. This research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors. The results of the current study do not constitute endorsement of the product by the authors or the journal.

References

1. Ajan T, Adamfi A, Chinen R, et al. *Technical and Competition Rules and Regulations*. Budapest, Hungary: International Weightlifting Federation, 2018.
2. Askow AT, Stone JD, Arndts DJ, et al. Validity and reliability of a commercially-available velocity and power testing device. *Sports* 6: 170, 2018.
3. Balsalobre-Fernández C, Marchante D, Baz-Valle E, et al. Analysis of wearable and smartphone-based technologies for the measurement of barbell velocity in different resistance training exercises. *Front Physiol* 8: 649, 2017.
4. Boulosa DA, Abreu L, Beltrame LG, Behm DG. The acute effect of different half squat set configurations on jump potentiation. *J Strength Cond Res* 27: 2059–2066, 2013.
5. Claudino JG, Cronin J, Mezêncio B, et al. The countermovement jump to monitor neuromuscular status: A meta-analysis. *J Sci Med Sport* 20: 397–402, 2017.
6. Coe R. *Effect Size Calculator*: Centre for Evaluation & Monitoring, 2002. Available at: <https://www.cem.org/effect-size-calculator>. Accessed July 2018.
7. Comfort P. Within- and between-session reliability of power, force, and rate of force development during the power clean. *J Strength Cond Res* 27: 1210–1214, 2013.
8. Comfort P. *A Biomechanical Analysis of Variations of the Power Clean and Their Application for Thletic Development*. Salford, UK: University of Salford, 2015.
9. Cormie P, McGuigan MR, Newton RU. Developing maximal neuromuscular power: Part 2-training considerations for improving maximal power production. *Sports Med* 41: 125, 2011.
10. Faigenbaum AD, McFarland JE, Herman R, et al. Reliability of the one repetition-maximum power clean test in adolescent athletes. *J Strength Cond Res* 26: 432, 2012.
11. Hackett D, Davies T, Soomro N, Halaki M. Olympic weightlifting training improves vertical jump height in sportspeople: A systematic review with meta-analysis. *Br J Sports Med* 50: 865–872, 2016.
12. Haff GG, Whitley A, McCoy LB, et al. Effects of different set configurations on barbell velocity and displacement during a clean pull. *J Strength Cond Res* 17: 95–103, 2003.
13. Hardee JP, Triplett NT, Utter AC, Zwetsloot KA, McBride JM. Effect of inter-repetition rest on power output in the power clean. *J Strength Cond Res* 26: 883–889, 2012.
14. Hopkins WG. Spreadsheets for analysis of validity and reliability. *Sportscience* 19: 36–45, 2015.

15. Kawamori N, Crum AJ, Blumert PA, et al. Influence of different relative intensities on power output during the hang power clean: Identification of the optimal load. *J Strength Cond Res* 19: 698–708, 2005.
16. Keppel G, Wickens TD. Simultaneous comparisons and the control of type I errors. In: *Design and Analysis: A Researcher's Handbook*. Upper Saddle River, NJ: Pearson Prentice Hall, 2004. pp. 111–130.
17. Kilduff LP, Bevan H, Owen N, et al. Optimal loading for peak power output during the hang power clean in professional rugby players. *Int J Sports Physiol Perform* 2: 260–269, 2007.
18. Kipp K, Meinerz C. A biomechanical comparison of successful and unsuccessful power clean attempts. *Sports Biomechanics* 16: 272–282, 2017.
19. Latella C, Teo WP, Drinkwater EJ, Kendall K, Haff GG. The acute neuromuscular responses to cluster set resistance training: A systematic review and meta-analysis. *Sports Med* 49: 1861–1877, 2019.
20. McBride JM, Haines TL, Kirby TJ. Effect of loading on peak power of the bar, body, and system during power cleans, squats, and jump squats. *J Sports Sci* 29: 1215–1221, 2011.
21. McBride JM, Triplett-McBride T, Davie A, Newton RU. A comparison of strength and power characteristics between power lifters, Olympic lifters, and sprinters. *J Strength Cond Res* 13: 58–66, 1999.
22. Merrigan JJ, Tufano JJ, Oliver JM, et al. Reducing the loss of velocity and power in women athletes via rest redistribution. *Int J Sports Physiol Perform* 15: 255–261, 2020.
23. Oliver JM, Kreutzer A, Jenke SC, et al. Velocity drives greater power observed during back squat using cluster sets. *J Strength Cond Res* 30: 235–243, 2016.
24. Seimetz C, Tan D, Katayama R, Lockhart T. A comparison between methods of measuring postural stability: Force plates versus accelerometers. *Biomed Sci Instrum* 48: 386–392, 2012.
25. Seitz LB, de Villarreal ES, Haff GG. The temporal profile of postactivation potentiation is related to strength level. *J Strength Cond Res* 28: 706–715, 2014.
26. Seitz LB, Haff GG. Factors modulating post-activation potentiation of jump, sprint, throw, and upper-body ballistic performances: A systematic review with meta-analysis. *Sports Med* 46: 231–240, 2016.
27. Suchomel TJ, Nimphius S, Bellon CR, Stone MH. The importance of muscular strength: Training considerations. *Sports Med* 48: 765–785, 2018.
28. Tillin N, Bishop D. Factors modulating post-activation potentiation and its affect on performance of subsequent explosive activities. *Sports Med* 39: 147–166, 2009.
29. Tufano JJ, Brown LE, Gregory Haff GG. Theoretical and practical aspects of different cluster set structures: A systematic review. *J Strength Cond Res* 31: 848–867, 2017.
30. Tufano JJ, Conlon JA, Nimphius S, et al. Maintenance of velocity and power with cluster sets during high-volume back squats. *Int J Sports Physiol Perform* 11: 885–892, 2016.
31. Turki O, Chaouachi A, Drinkwater EJ, et al. Ten minutes of dynamic stretching is sufficient to potentiate vertical jump performance characteristics. *J Strength Cond Res* 25: 2453–2463, 2011.
32. Wadhi T, Rauch JT, Tamulevicius N, Andersen JC, De Souza EO. Validity and reliability of the gymaware linear position transducer for squat jump and counter-movement jump height. *Sports* 6: 177, 2018.
33. Young WB, Pryor JF, Wilson GJ. Countermovement and drop jump performance. *J Strength Cond Res* 9: 232–236, 1995.